

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2082 Shrawan

Exam.	Back		
Level	BE	Full Marks	80
Programme	BEL, BEX BEL, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electromagnetics (EX 503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks
- ✓ Necessary formula are attached herewith.
- ✓ Assume suitable data if necessary.

- Transform the vector $\vec{A} = 4\hat{a}_x - 2\hat{a}_y - 4\hat{a}_z$ into spherical co-ordinates at a point $P(x = -2, y = -3, z = 4)$. [5]
- An infinite uniform line charge $\rho_L = 2\text{nC/m}$ lies along the x-axis in free space, while point charges of 8nC each are located at $(0, 0, 1)$ and $(0, 0, -1)$. Find $\vec{D}$ at $(2, 3, -4)$. [5]
- State and prove uniqueness theorem. [2+6]
- Using the continuity equation, elaborate the concept of Relaxation Time Constant (RTC) with necessary derivations. Let $\vec{J} = \frac{e^{-10^4 t}}{\rho^2} \hat{a}_\rho$ A/m² be the current density in a given region. At $t = 10\text{ms}$, calculate the amount of current passing through surface $\rho = 2\text{m}$, $0 \leq z \leq 3\text{m}$, $0 \leq \phi \leq 2\pi$. [4+4]
- State and prove the Stoke's Theorem. Calculate the value of the vector current density in cylindrical coordinates at $P_B(1.5, 90^\circ, 0.5)$ if $\vec{H} = \frac{2}{\rho}(\cos 0.2\phi) \hat{a}_\rho$. [5+3]
- Define scalar magnetic potential. The region $y < 0$ (region 1) is air and $y > 0$ (region 2) has $\mu_r = 10$. If there is a uniform magnetic field $\vec{H} = 5\hat{a}_x + 6\hat{a}_y + 7\hat{a}_z$ A/m in region 2, find $\vec{B}$ and $\vec{H}$ in region 2. [2+6]
- List out the Maxwell equations phasor form for time varying case in free space. Derive the wave equation for the wave propagating in Lossy Media. [2+6]
- A uniform plane wave in free space is given by $\vec{H}_S = (250 \angle 30^\circ) e^{-j350z} \hat{a}_x$ V/m. Determine phase constant, frequency of the wave, intrinsic impedance, $\vec{E}_S$ and the magnitude $\vec{H}$ of at $z = 25\text{mm}$ and $t = 4\text{ps}$. [1+2+1+2+2]
- Within a certain region, $\epsilon = 10^{-11}\text{F/m}$ and $\mu = 10^{-5}\text{H/m}$. If $B_x = 2 \times 10^{-4} \cos 10^5 t \sin 10^{-3} y$ T find: [3+3+2]
 - Find $\vec{E}$
 - Find the total magnetic flux passing through the surface $x = 0$, $0 < y < 40\text{ m}$, $0 < z < 2\text{ m}$ at $t = 1\text{ }\mu\text{s}$
 - Find the value of the closed line integral of $\vec{E}$ around the perimeter of the given surface.

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Subject: - Electro-magnetics (EX 503)

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1. Express the vector field, $D = (x^2 + y^2)^{-1} (x\mathbf{a}_x + y\mathbf{a}_y)$ in cylindrical system and find D at $\rho = 2$, $\phi = 0.2\pi$, and $z = 5$. [5]
2. Derive the expression for electric field intensity of an infinite line charge having uniform line charge density of ρ_{LC}/m using Coulomb's Law. [6]
3. Let a uniform line charge density of 8 nC be located at $x=0, z=4$ and a point charge of 2 μ C be present at P(2,0,0). If $V = 0$ at M (0,0,5) find V at N(1,2,3). [5]
4. Find the expression for energy stored in an Electric field. [5]
5. State Biot-Savart's Law. Find the magnetic field intensity at point P (2, 3, 5) if there is an infinitely long straight current carrying filament passing through the origin and point Q (0, 1, 0). The current of 50 A is directed from origin to point Q. [6]
6. Verify stoke's theorem for the field $\vec{H} = (\frac{3r^2}{\sin\theta})\vec{a}_\theta + 54r\cos\theta\vec{a}_\phi$ A/m in free space for the conical surface defined by $\theta = 20^\circ$, $0 \leq \phi \leq 2\pi$, $0 \leq r \leq 5$. Let the positive direction of $\vec{ds}$ be $\vec{a}_\theta$. [5]
7. Derive an expression for Torque on a closed circuit. [6]
8. Derive the expressions for electric field and magnetic field for a uniform plane wave propagating in free space. Also prove that the uniform plane wave propagates with the speed of light in the free space. [8]
9. An EM wave travels in free space with the electric field component $\vec{E} = (15\vec{a}_y - 5\vec{a}_x)\cos(\omega t - 3y + 5z)$ V/m. Find; [8]
 - a) ω and λ .
 - b) The magnetic field component.
10. The region $z < 0$ contains a perfect dielectric for which $\epsilon_{r1} = 2.5$, while the region $z > 0$ is characterized by $\epsilon_{r2} = 4$. Let $\vec{E}_1 = -30\hat{a}_x + 50\hat{a}_y + 70\hat{a}_z$ V/m, find the electric field intensities, electric flux densities and polarization in region 2. [6]
11. Find Capacitance for a capacitor with two layer dielectric. [5]
12. State Faraday's Law. Correct the Ampere's Circuital Law with necessary arguments and derivation for time varying fields. [5]
13. A transmission line operating at 200 MHz has $R = 20 \Omega/m$, $L = 0.3 \mu H/m$, $C = 63 \rho F/m$ and $G = 4.2 \text{ ms/m}$. Find; [7]
 - a) Propagation coefficient (γ).
 - b) Phase constant and attenuation constant.
 - c) Characteristics impedance (Z_0).
 - d) Velocity of wave propagation on the line (v).
14. What is a waveguide? Explain its advantages over transmission line. [3]

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1. At point P(-3, -4, 5), express that vector that extends from P to Q(2, 0, -1) in spherical coordinates. [5]
2. Find $\vec{D}$ at P (6, 8, 10) caused by a point charge of 30 nC at the origin, a uniform line charge $\rho_L = 40 \mu\text{C/m}$ on the z-axis and a uniform surface charge density $\rho_s = 57.2 \mu\text{C/m}^2$ on the plane $x = 9$. [6]
3. Derive the expression for the Maxwell's equation in point form. [6]
4. Find the equation of a streamline that passes through the point P(1,4,-2) in the field $E = -8x/y \mathbf{a}_x + 4x^2/y^2 \mathbf{a}_y$. [4]
5. Derive an expression for the magnetic field intensity produced by an infinitely long filament carrying current using Biot and Savart Law. [6]
6. What is curl? State and Prove Stokes theorem. [5]
7. A square loop of wire in the $z = 0$ plane with co-ordinates (1,0,0) (1,2,0) (3,2,0) and (3,0,0) carrying 2mA is placed in the field of an infinite filament on the y axis with current of 15A in $-\mathbf{a}_y$ direction. Determine the total force on the loop. [6]
8. Derive an expression for standing Wave for both electric and magnetic field. Also indicate where on the z-axis you'll get the maximum and minimum value of electric field intensity E. Assume that the boundary is at $z = 0$, the region $z < 0$ is a perfect dielectric and the region $z > 0$ may be of any material. [8]
9. A 1 MHz uniform plane wave with an amplitude of 25 V/m is propagating along $\hat{\mathbf{a}}_x$ direction in a material for which $\epsilon_r = 4$, $\mu_r = 9$ and $\sigma = 0$. Find: [8]
 - a) The velocity of propagation.
 - b) The phase constant.
 - c) The intrinsic impedance.
 - d) $\vec{E}(t)$ if $E_z = 0$ and $E_y = 25 \text{ V/m}$ at P (10, 10, 10) at 100 ns.
 - e) $\vec{H}(t)$
10. Use boundary condition to find $\vec{E}_2$ the medium 2 with boundary located at plane $z=0$. Medium 1 is perfect dielectric characterized by $\epsilon_{r1}=2.5$, medium 2 is perfect dielectric characterized by $\epsilon_{r2}=5$, electric field in medium is $\vec{E}_1=\vec{a}_x+3\vec{a}_y+3\vec{a}_z \text{ v/m}$. Also find the angles made by electric field $\vec{E}_2$ with the boundary interface. [6]
11. Find Capacitance per unit length of a co-axial cable using Laplace equation. [5]
12. Explain Faradays law. Derive the relation of the motional emf and displacement current. [5]
13. The parameters of a certain transmission line operating at $6 \times 10^8 \text{ rad/s}$ are $L = 0.4 \mu\text{H/m}$, $C = 40 \text{ pF/m}$, $G = 80 \text{ mS/m}$, and $R = 20 \Omega/\text{m}$. Find γ , α , β , λ , and Z_0 . [7]
14. Write short notes on types of antenna and antenna parameters. [3]

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- Express a scalar potential field $V = x^2 + 2y^2 + 3z^2$ in spherical coordinates. Find the value of P at a point $(2, 60^\circ, 90^\circ)$. [4+1]
- Derive the expression for an electric field intensity due to an infinite line charge using Gauss's Law. Find electric flux density at point $P(6, 5, 4)$ due to a uniform line charge of 6 nC/m at $x = 4$, $y = 2$, point charge 10 nC at $Q(3, 2, 4)$ and uniform surface charge density of 0.4 nC/m^2 at $x = 3$. [3+5]
- Two uniform charges 8 nC/m are located at $x = 1, z = 2$ and $x = -1, y = 2$ in free space respectively. If the potential at origin is 100 V . Find V at $P(4, 1, 3)$. [6]
- Derive Poisson Equation. Find the capacitance of parallel plate capacitor by solving Laplace equation with potential difference between the plates as V_0 . [3+4]
- Evaluate both sides of Stokes's theorem for the field $\vec{H} = 12 \sin\theta \hat{a}_\theta$ and the surface $r = 4, 0 \leq \theta \leq 90^\circ, 0 \leq \phi \leq 90^\circ$. Let the surface have the $\hat{a}_r$ direction [6]
- Differentiate Scalar Magnetic Potential and Vector Magnetic Potential. Given magnetic vector potential $\vec{A} = -\frac{\rho^2}{4} \hat{a}_z\text{ Wb/m}$. Calculate the total magnetic flux crossing the surface $\phi = \frac{\pi}{2}, 1 \leq \rho \leq 2, 0 \leq z \leq 5$. [2+6]
- List out point form of Maxwell's equations in phasor form for time varying case. Using these equations, derive the electric field component of a uniform plane wave travelling in the perfect dielectric medium. [2+6]
- A 9.375 GHz uniform plane wave is propagating in polythene ($\epsilon_r = 2.26, \mu_r = 1$). If the amplitude of the electric field intensity is 500 V/m and the material is assumed to be lossless: Find the phase constant, wavelength, velocity of propagation and intrinsic impedance. [2+2+2+2]
- Explain the term Skin Depth. Using Poynting Vector deduce the time- average power density for a perfect dielectric. [4+4]
- A lossless $60\ \Omega$ line is 1.8λ long and is terminated with pure resistance of $80\ \Omega$. The load voltage is $15\angle 30^\circ\text{ V}$. (i) Average power delivered to load (ii) Magnitude of minimum voltage on the line. [4+4]
- What are the advantages of waveguides over transmission line? Consider a rectangular waveguide of dimension $a = 1.07\text{ cm}$, $b = 0.43\text{ cm}$. Find the cutoff frequency for TM_{11} mode. ($E_r = 2, \mu = \mu_0$) [2+4]
- Write short notes about antenna and its parameters. [2]

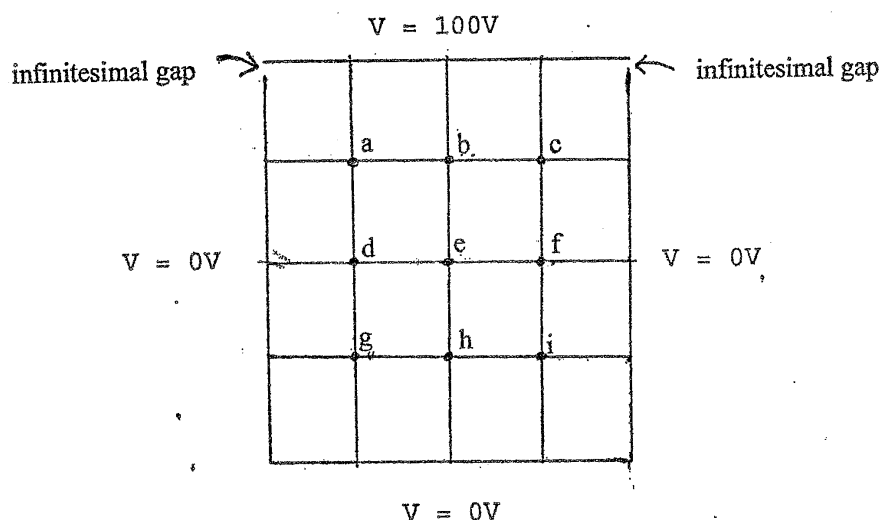
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- The Magnetic field Intensity in a certain region is given as $\vec{H} = 20\hat{a}_\rho - 10\hat{a}_\phi + 3\hat{a}_z$ A/m. Transform this field vector into Cartesian co-ordinate at $P(x=5, y=2, z=-1)$. [5]
- A point charge of $6\mu\text{C}$ located at origin, uniform line charge density of 180nc/m lies along x-axis and uniform sheet charge of 25c/m^2 lies on $z=0$ plane. Find $\vec{D}$ at point $(1,2,4)$. [6]
- Differentiate between divergence and gradient. Let $V = \frac{\cos 2\phi}{\rho}$ in free space. (i) Find the volume charge density at point A $(0.5, 60^\circ, 1)$. (ii) Find the surface charge density on a conductor surface passing through the point B $(2, 30^\circ, 1)$. [2+3+3]
- Define curvilinear square method for calculating capacitance. A square grid shown in figure below within a given potential. Find the potential at point a, b, c, d, e, f, g, h and i using iteration method. Complete it using single iteration only. [4+4]



- Justify the maxwell's equation: $\oint \vec{B} \cdot d\vec{S} = 0$ with necessary remarks. Derive an expression of magnetic field intensity for an infinite filament carrying a direct current using vector magnetic potential. [2+6]

6. Define Biot savart's law. Let the permittivity be $5\mu\text{H/m}$ in region A where $x < 0$, and $20\mu\text{H/m}$ in region B where $x > 0$. If there is a surface current density ,
 $\vec{K} = 150\hat{a}_y - 200\hat{a}_z \text{ A/m}$ at $x = 0$ and if $\vec{H} = 300\hat{a}_x - 400\hat{a}_y + 500\hat{a}_z \text{ A/m}$, find
 $|\vec{H}_{tA}|$, $|\vec{H}_{nA}|$, $|\vec{H}_{tB}|$ and $|\vec{H}_{nB}|$ [2+6]
7. Define poynting vector. Using this deduce the time average power density for a dissipative medium. [2+6]
8. A conductor with cross-sectional area of 10cm^2 carries conduction current
 $\vec{J} = 0.2\sin 10^9 t \hat{a}_z \text{ mA}$. Given that $\sigma = 2.5 \times 10^6 \text{ S/m}$, and $\epsilon_r = 6$. Calculate the magnitude
of the displacement current density. [5]
9. A uniform plane wave in free space is given by Magnetic field intensity $\vec{H}$ in phasor
form as: $\vec{H}_s = 400 \angle 30^\circ e^{-j250z} \hat{a}_y \text{ A/m}$ Find:
a) Angular frequency (ω).
b) Wavelength (λ) and intrinsic impedance (η).
c) Electric field intensity $\vec{E}(x, y, z, t)$ at $z = 50\text{mm}$ and $t = 4\text{pS}$. [2+2+4]
10. The parameters of a certain transmission line operating at $6 \times 10^8 \text{ rad/s}$ are $L = 0.4\mu\text{H/m}$,
 $C = 40\text{pF/m}$, $G = 80\text{mS/m}$, and $R = 20\Omega/\text{m}$. Find γ , λ and Z_0 (characteristic impedance). [3+3+2]
11. A standard air-filled rectangular waveguide with dimensions $8.636\text{cm} \times 4.318\text{cm}$ is fed by
a 4GHz carrier from a coaxial cable. Determine if a TE_{10} mode will be propagating or not. [6]
12. What are the parameters of antenna? List out the types of Antenna. [2]

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- Transform the vector $\vec{A} = y\vec{a}_x - (x+y)\vec{a}_y + z\vec{a}_z$ at point P (-3,4,5) in cylindrical coordinate system. [5]
- Define electric flux density. Given the field $\vec{D} = \frac{20}{\rho^2}(-\sin^2 \phi \hat{a}_\rho + \sin 2\phi \hat{a}_\phi)$, evaluate both sides of the divergence theorem for the region bounded by $1 < \rho < 2, 0 < \phi < 90^\circ, 0 < z < 1$. [1+7]
- State Divergence's Theorem: A current density in certain region is given as
$$\vec{j} = \frac{400 \sin \theta \cdot \vec{a}_r}{r^2} \text{ A/m}^2.$$
 Find the total current flowing through that portion of the spherical surface $r = 0.8$ bounded by $0.1\pi < \theta < 0.3\pi, 0 < \phi < 2\pi$. [2+6]
- Find the equation for Energy Density in the electrostatic field. [7]
- Define curl. Evaluate both side of Stoke's Theorem for $\vec{H} = 8z\vec{a}_x - 4x^3\vec{a}_z$ A/m and rectangular path P(2,3,4) to Q(4,3,4) to R(2,3,1) to S(2,3,1) to P. [1+7]
- State Biot-Savart's law. A filamentary current of 10A is directed in from infinity to the origin on the positive x-axis and then block out to infinity along the positive z-axis. Use the Biot-Savart's law to determine $\vec{H}$ at P(0,1,0). [2+6]
- State Faraday's law of electromagnetic induction. Explain motional induction and transformer induction with necessary expressions. [1+3+3]
- Derive an expression for electric field and magnetic field for a uniform plane wave propagating in a free space. [7]
- Determine skin depth, propagation constant and velocity of wave at 1 MHz in good conductor with conductivity of 1.9×10^7 mho per meter. [2+2+2]
- A 200Ω transmission line is lossless, 0.25λ long and is terminated in $Z_L = 400\Omega$. The line has the generator with $80 \angle 0^\circ$ V in series with 100Ω connected to the input. (a) Find the load voltage (b) Find the voltage at the midpoint of the line. [4+4]
- Explain the different modes of propagation supported by waveguides. A rectangular waveguide has a cross-section of $2.5 \text{ cm} \times 1.2 \text{ cm}$. Determine if the signal of 5 GHz propagates in the dominant mode. [2+4]
- What are the parameters of antenna? List out the types of Antenna. [2]

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- ✓ $\vec{A}$ represent a vector and $\hat{a}_{\text{subscript}}$ denotes a unit vector along the direction given by the subscript.
- ✓ Assume suitable data if necessary.

- Transform the vector $\vec{F}$ into the cylindrical co-ordinate system.
 $\vec{F} = 10\hat{a}_x - 8\hat{a}_y + 6\hat{a}_z$ at point $P(x = 10, y = -8, z = 6)$ [5]
- Define electric dipole moment. Two uniform line charges, 8 nC/m each, are located at $x = 1, z = 2$ and at $x = -1, y = 2$ in free space. If the potential at the origin is 100 V, find V at $P(4, 1, 3)$. [2+6]
- State Gauss's Law. The region $y < 0$ contains a dielectric material for which $\epsilon_{r1} = 2.5$, while the region $y > 0$ is characterized by $\epsilon_{r2} = 4$. Let $\vec{E}_1 = -30\hat{a}_x + 50\hat{a}_y + 70\hat{a}_z$ V/m, find the electric field intensities, flux densities in region 2 and the angle θ_1 , which is the angle made by normal component of $\vec{E}$ or $\vec{D}$ with total $\vec{E}$ or $\vec{D}$. [2+3+2+1]
- Derive Poisson's equation. Assuming that the potential V in the cylindrical coordinate system is the function of ρ only, solve the Laplacian equation by integration method and derive the expression for the capacitance of co-axial capacitor using the same solution of V. [2+5]
- State Stoke's theorem. Evaluate both sides of Stoke's theorem for the field $\vec{H} = 8xy\hat{a}_x - 5y^2\hat{a}_y$ A/m and the rectangular path around the region $2 \leq x \leq 5, -1 \leq y \leq 1, z = 0$. Let the positive direction of $d\vec{S}$ be $\hat{a}_z$. [1+7]
- Define Ampere's Circuital law. Determine H at $P_2(0.4, 0.3, 0)$ in the field of an 8 A filamentary current directed inward from infinity to the origin on the positive x axis, and then outward to infinity along the y axis. [2+6]
- Explain motional induction with necessary derivations. Correct the equation $\nabla \times \vec{H} = \vec{J}$ with necessary arguments and derivation for time varying fields. [3+4]
- Derive the expression for electric and magnetic fields for a uniform plane wave propagating in a dissipative medium. [4+3]
- A uniform plane wave in free space is given by $\vec{H}_s = (250 \angle 30^\circ)e^{-j350z}\hat{a}_x$ V/m. Determine phase constant, frequency of the wave, intrinsic impedance, $\vec{E}_s$ at $z = 25$ mm and $t = 4$ ps. [1+2+1+2]
- Define the secondary parameters of a transmission line. A lossless transmission line with $Z_0 = 50$ ohm has a length of 0.4λ . The operating frequency is 300 MHz and it is terminated with a load $Z_L = 40 + j30$. Find: [2+1+2+3]
 - Reflection Coefficient
 - Standing wave ratio on the line (SWR)
 - Input impedance (Z_{in})
- Differentiate between TE and TM modes. Consider a rectangular waveguide with $\epsilon_r = 4, \mu = \mu_0$ with dimensions $a = 2.08$ cm, $b = 0.54$ cm. Find the cutoff frequency for TM_{11} mode and the dominant mode. [3+3]
- Write short note on antenna and its types. [2]

Divergence

$$\text{Cartesian: } \nabla \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\text{Cylindrical: } \nabla \cdot \vec{D} = \frac{1}{\rho} \frac{\partial(\rho D_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

$$\text{Spherical: } \nabla \cdot \vec{D} = \frac{1}{r^2} \frac{\partial(r^2 D_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(D_\theta \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi}$$

Gradient

$$\text{Cartesian: } \nabla V = \frac{\partial V}{\partial x} \hat{a}_x + \frac{\partial V}{\partial y} \hat{a}_y + \frac{\partial V}{\partial z} \hat{a}_z$$

$$\text{Cylindrical: } \nabla V = \frac{\partial V}{\partial \rho} \hat{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \hat{a}_\phi + \frac{\partial V}{\partial z} \hat{a}_z$$

$$\text{Spherical: } \nabla V = \frac{\partial V}{\partial r} \hat{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{a}_\phi$$

Curl

$$\text{Cartesian: } \nabla \times \vec{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \hat{a}_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \hat{a}_y + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \hat{a}_z$$

$$\text{Cylindrical: } \nabla \times \vec{H} = \left(\frac{1}{\rho} \frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z} \right) \hat{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho} \right) \hat{a}_\phi + \frac{1}{\rho} \left(\frac{\partial(\rho H_\phi)}{\partial \rho} - \frac{\partial H_\rho}{\partial \phi} \right) \hat{a}_z$$

$$\text{Spherical: } \nabla \times \vec{H} = \frac{1}{r \sin \theta} \left(\frac{\partial(H_\phi \sin \theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi} \right) \hat{a}_r + \frac{1}{r} \left(\frac{\partial H_r}{\partial \sin \theta} - \frac{\partial(r H_\theta)}{\partial r} \right) \hat{a}_\theta + \frac{1}{r} \left(\frac{\partial(r H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta} \right) \hat{a}_\phi$$

Laplacian:

$$\text{Cartesian: } \nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\text{Cylindrical: } \nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\text{Spherical: } \nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

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INSTITUTE OF ENGINEERING
Examination Control Division
2079 Baishakh

Exam. Level	Back		
	BE	Full Marks	80
Programme	BEL, BEX, BEL, BCT	Pass Marks	32
	Year / Part	Time	3 hrs.
	II / I		

Subject: - Electromagnetics (EX 503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Assume suitable data if necessary.

- Express the vector field $A = (x-y) \hat{a}_y$ in cylindrical and spherical coordinate systems. [5]
- Find the total charge inside the volume indicated:

$$\rho_v = 4xyz^2, 0 \leq \rho \leq 2, 0 \leq \phi \leq \frac{\pi}{2}, 0 \leq z \leq 3. \quad [4]$$

- Obtain the equation of the streamline that passes through the point P (-2,7,10) in the field:

$$\vec{E} = 2(y-1)\hat{a}_x + 2x\hat{a}_y. \quad [4]$$

- Given the potential field in cylindrical coordinates, $V = [100/(z^2 + 1)] \rho \cos\phi$ V, and point P at $\rho = 3$ m, $\phi = 60^\circ$, $z = 2$ m, find values at P for (a) V; (b) E; (c) $\vec{E}$; (d) dV/dN ; (e) a_N ; (f) ρ_v in free space. [6]

- Define gradient and laplacian function. A point charge of 16nC is located at Q (2,3,5) in free space and a uniform line charge of 5nC/m is at the intersection of the plane $x=2$ and $y=4$. If the potential at the origin is 100V, Find V at P (4,1,3). [2+6]

- Define curl and its significance in Electromagnetics. Evaluate both sides of stokes theorem for the field $H = 6xy \hat{a}_x - 3y^2 \hat{a}_y$ A/m and the rectangular path around the region $2 \leq x \leq 5, -1 \leq y \leq 1, z = 0$, let the positive direction of ds be $\hat{a}_z$. [2+6]

- Justify the Maxwell's equation: $\oint_S \vec{B} \cdot d\vec{S} = 0$ with necessary remarks. Derive an expression of magnetic field intensity for an infinite filament carrying a direct current using vector magnetic potential. [2+6]

- Write down the Maxwell equation in phasor form. Derive the equation for electric field for a uniform plane wave travelling in air. [2+6]

- A uniform plane wave in free space is given by Electric field intensity $\vec{E}$ in phasor form as:

$$\vec{E}_s = 200 \angle 30^\circ e^{-j250z} \hat{a}_x \text{ V/m Find:}$$

- Angular frequency (ω)
- Wavelength (λ) and intrinsic impedance (η)
- Magnetic field intensity $\vec{H}(x,y,z,t)$ at $z = 8$ mm and $t = 6$ pS. [2+2+4]

10. Define Faraday's law. A conductor with cross-sectional area of 10 cm^2 carries conduction current $\vec{J} = 0.2 \sin 10^9 t \hat{a}_z \text{ mA}$. Given that $\sigma = 2.5 \times 10^6 \text{ S/m}$, and $\epsilon_r = 6$. Calculate the value of the displacement current. [2+4]
11. A lossless transmission line is 80 cm long and operates at a frequency of 600 MHz. The line parameters are $L = 0.25 \mu\text{H/m}$ and $C = 100 \text{ pF/m}$. Find the characteristic impedance, the phase constant, the phase velocity on the line, and the input impedance for $Z_L = 100 \Omega$. [8]
12. Define dominant mode. A standard air-filled rectangular waveguide with dimensions $8.636 \text{ cm} \times 4.318 \text{ cm}$ is fed by a 8 GHz carrier from a coaxial cable. Determine if a TE_{10} mode will be propagating or not. [1+4]
13. Write short notes on antenna and its parameters. [2]

TRIBHUVAN UNIVERSITY
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Examination Control Division
2078 Kartik

Exam.	Back		
Level	BE	Full Marks	80
Programme	BEL, BEX, BEL, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electromagnetics (EX 503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary data are attached herewith.
- ✓ $\vec{A}$ represent a vector and $\vec{a}_{\text{subscript}}$ denotes a unit vector along the direction given by the subscript.
- ✓ Assume suitable data if necessary.

1. Transform a vector field $\vec{A} = 4\vec{a}_x - 2\vec{a}_y - 4\vec{a}_z$ into cylindrical coordinate system at a point P (2, 3, 5). [5]
2. A plane $x = 2$ carry a surface charge density 10 nC/m^2 , a line $x = 0$ and $z = 3$ carry a line charge density 10 nC/m and a point charge of 10 nC is at origin. Calculate $\vec{E}$ at (1, 1, -1) due to these charge configurations. [7]
3. Evaluate the both sides of divergence theorem for the field $\vec{D} = 2xy\vec{a}_x + x^2\vec{a}_y \text{ C/m}^2$ and the rectangular parallelopiped formed by the planes $x = 0$ and 1, $y = 0$ and 2, and $z = 0$ and 3. [7]
4. If potential field in free space is $V = \frac{10}{r^2} \sin\theta \cos\phi \text{ V}$ and point P is located at (2, 90° , 0°). Find: (a) $\vec{E}$ (b) direction of $\vec{E}$ at P (c) energy density at P. [2+2+2]
5. Find the vector magnetic field intensity $\vec{H}$ in Cartesian coordinates at P (2, 1, 3) caused filament of 12 Ampere(A) in a $\vec{a}_z$ direction on the z-axis and extending from $z = 0$ to $z = 4$. [8]
6. Consider a boundary at $z = 0$ which carries current $\vec{K} = \left(\frac{1}{\mu_0}\right)\vec{a}_y \text{ mA/m}$. Medium 1 ($z < 0$) is filled with material whose $\mu_r = 6$ and medium 2 ($z > 0$) is filled with material whose $\mu_r = 4$. If $\vec{B}_2 = 5\vec{a}_x + 8\vec{a}_z \text{ mT}$, find $\vec{B}_1$. [8]
7. Define Poynting vector. Using this deduce the time average power density for a dissipative medium. [2+5]
8. A uniform plane wave has a magnetic field component $\vec{H} = 15 \cos(2 \times 10^8 t + \beta x) \vec{a}_y \text{ A/m}$ in a medium characterized by $\sigma = 0$, $\epsilon = 4\epsilon_0$, $\mu = \mu_0$. Find [5+1+2]
 - a) direction of propagation, phase constant β , wavelength λ , velocity v_p , intrinsic impedance η
 - b) Magnitude of $\vec{H}$
 - c) $\vec{E}$
9. A uniform plane wave in air partially reflects from the surface of a material whose properties are unknown. Measurements of the electric field in the region in front of the interface yield a 1.5 m spacing between maxima, with the first maximum occurring 0.75 m from the interface. A standing wave ratio (SWR) of 5 is measured. Determine the intrinsic impedance of the unknown material. [8]

10. A 50Ω lossless transmission line is 0.4λ long. The line is terminated with a load $Z_L = 40 + j30 \Omega$. If the operating frequency is 300 MHz, find [2+2+4]
- reflection coefficient (Γ)
 - standing wave ratio (s) and
 - input impedance (Z_m)
11. Explain why TEM wave doesn't exist in a rectangular waveguide? A rectangular waveguide has dimensions $a = 1 \text{ cm}$, $b = 2 \text{ cm}$. The medium within the waveguides has $\epsilon_r = 1$, $\mu_r = 1$, $\sigma = 1$. Find whether or not the signal with the frequency of 500 MHz will be transmitted in the $TE_{1,0}$ mode. [2+4]
12. What are the parameters of antenna? List out the different types of antenna you have studied. [1+1]

Divergence

$$\text{Cartesian: } \nabla \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\text{Cylindrical: } \nabla \cdot \vec{D} = \frac{1}{\rho} \frac{\partial(\rho D_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

$$\text{Spherical: } \nabla \cdot \vec{D} = \frac{1}{r^2} \frac{\partial(r^2 D_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(D_\theta \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi}$$

Gradient

$$\text{Cartesian: } \nabla V = \frac{\partial V}{\partial x} \hat{a}_x + \frac{\partial V}{\partial y} \hat{a}_y + \frac{\partial V}{\partial z} \hat{a}_z$$

$$\text{Cylindrical: } \nabla V = \frac{\partial V}{\partial \rho} \hat{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \hat{a}_\phi + \frac{\partial V}{\partial z} \hat{a}_z$$

$$\text{Spherical: } \nabla V = \frac{\partial V}{\partial r} \hat{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{a}_\phi$$

Curl

$$\text{Cartesian: } \nabla \times \vec{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \hat{a}_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \hat{a}_y + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \hat{a}_z$$

$$\text{Cylindrical: } \nabla \times \vec{H} = \left(\frac{1}{\rho} \frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z} \right) \hat{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho} \right) \hat{a}_\phi + \frac{1}{\rho} \left(\frac{\partial(\rho H_\phi)}{\partial \rho} - \frac{\partial H_\rho}{\partial \phi} \right) \hat{a}_z$$

$$\text{Spherical: } \nabla \times \vec{H} = \frac{1}{r \sin \theta} \left(\frac{\partial(H_\phi \sin \theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi} \right) \hat{a}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial H_r}{\partial \phi} - \frac{\partial(r H_\phi)}{\partial r} \right) \hat{a}_\theta + \frac{1}{r} \left(\frac{\partial(r H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta} \right) \hat{a}_\phi$$

Laplacian:

$$\text{Cartesian: } \nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\text{Cylindrical: } \nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\text{Spherical: } \nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

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INSTITUTE OF ENGINEERING
Examination Control Division
2078 Bhadra

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEL, BEX, BEI, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electro-magnetics (EX 503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary data are attached herewith.
- ✓ $\vec{A}$ represent a vector and $\vec{a}_{\text{subscript}}$ denotes a unit vector along the direction given by the subscript.
- ✓ Assume suitable data if necessary.

- Given a point P (-2, 6, 3) and vector field $\vec{A} = y\vec{a}_x + (xy + z)\vec{a}_y$, express P and $\vec{A}$ in spherical co-ordinate system. [5]
- A point charge of $6\mu\text{C}$ located at origin, uniform line charge density of 180nc/m lies along x-axis and uniform sheet charge of 25 C/m^2 lies on $z = 0$ plane. Find $\vec{D}$ at point (1, 2, 4). [7]
- Derive the expression for an electric field intensity due to an infinitely long line charge with charge density ρ_L by using Gauss's law. Find the volume charge density that is associated with the field $\vec{D} = xy^2\vec{a}_x + x^2y\vec{a}_y + z\vec{a}_z\text{ C/m}^2$. [4+3]
- State continuity equation. Given the vector current density $\vec{J} = 10\rho^2 z\vec{a}_\rho - 4\rho\sin^2\phi\vec{a}_\phi\text{ mA/m}^2$. Determine the current following outward the circular band $\rho = 5$, $0 < \phi < 2\pi$, $2 < z < 2.8$. [2+4]
- Differentiate between scalar magnetic potential and vector magnetic potential. If a vector magnetic potential is $\vec{A} = -(\rho^2/4)\vec{a}_z\text{ wb/m}$, calculate total magnetic flux crossing the surface $\phi = \pi/2$, $1 \leq \rho \leq 2\text{ m}$ and $0 \leq z \leq 5\text{ m}$. [4+4]
- The region $y < 0$ (region 1) is air and $y > 0$ (region 2) has $\mu_r = 10$. If there is a uniform magnetic field $\vec{H} = 5\vec{a}_x + 6\vec{a}_y + 7\vec{a}_z\text{ A/m}$ in region 1, find $\vec{B}$ and $\vec{H}$ in region 2. [8]
- Correct the equation $\nabla \times \vec{E} = 0$ for time varying field with necessary derivation. Also modify the equation $\nabla \times \vec{H} = \sigma\vec{E}$ with necessary arguments and derivation for time varying field. [3+4]
- A uniform plane wave in free space is given by $\vec{H}_s = (250\angle 30^\circ)e^{-j350z}\vec{a}_x\text{ V/m}$. Determine phase constant, frequency of the wave, intrinsic impedance, $\vec{E}_s$ and the magnitude $\vec{H}$ of at $z = 25\text{ mm}$ and $t = 4\text{ps}$. [1+1+2+2+2]
- Derive the expression for electric and magnetic fields for a uniform plane wave propagating in a free space. [8]
- A lossless transmission line is 80 cm long and operates at a frequency 1 GHz . The line parameters are $L = 0.5\text{ }\mu\text{H/m}$ and $C = 200\text{ pF/m}$. Find the characteristics impedance, the phase constant, the velocity on the line, and the input impedance for $Z_L = 100\Omega$. [2+2+2+2]
- Write short notes on TE and TM modes of rectangular waveguide. An air filled rectangular waveguide has cross-section of $2.3\text{ cm} \times 1.02\text{ cm}$. Calculate the cutoff frequency of the dominant mode (TE_{10}). [3+3]
- Write short notes about antenna and its parameters. [2]

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INSTITUTE OF ENGINEERING
Examination Control Division
2076 Chaitra

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEL, BEX, BEL, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electromagnetics (EX 503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formulas are attached herewith.
- ✓ Assume suitable data if necessary.

1. Transform the vector $\vec{A} = 4\hat{a}_x - 2\hat{a}_y - 4\hat{a}_z$ into spherical co-ordinates at a point P(x = -2, y = -3, z = 4). [5]
2. An infinite uniform line charge $\rho_L = 2\text{nC/m}$ lies along the x-axis in free space, while point charges of 8nC each are located at (0, 0, 1) and (0, 0, -1). (a) Find $\vec{D}$ at (2, 3, -4). [6]
3. Define uniqueness theorem. Find the energy stored in free space for the region $2\text{mm} < r < 3\text{mm}$, $0 < \theta < 90^\circ$, $0 < \phi < 90^\circ$, given the potential field $V =$: [2+6]
 - a) $\frac{200}{r}\text{V}$ and
 - b) $\frac{300}{r^2}\cos\theta\text{V}$
4. Using the continuity equation elaborate the concept of Relaxation Time Constant (RTC) with necessary derivations. Let $\vec{J} = \frac{e^{-10^{4t}}}{\rho^2} \hat{a}_\rho \text{ A/m}^2$ be the current density in a given region. At $t = 10\text{ms}$, calculate the amount of current passing through surface $\rho = 2\text{m}$, $0 \leq z \leq 3\text{m}$, $0 \leq \phi \leq 2\pi$. [4+4]
5. State and prove the Stoke's Theorem. Calculate the value of the vector current density: In cylindrical coordinates at $P_B(1.5, 90^\circ, 0.5)$ if $\vec{H} = \frac{2}{\rho}(\cos 0.2\phi) \hat{a}_\rho$. [5+3]
6. Define scalar magnetic potential. The region $y < 0$ (region 1) is air and $y > 0$ (region 2) has $\mu_r = 10$. If there is a uniform magnetic field $\vec{H} = 5\hat{a}_x + 6\hat{a}_y + 7\hat{a}_z \text{ A/m}$ in region 2, find $\vec{B}$ and $\vec{H}$ in region 2. [2+6]
7. List out the Maxwell equations phasor form for time varying case in free space. A conducting bar can slide freely over two conducting rails placed at $x = 0$ and $x = 10\text{cm}$. Calculate the induced voltage in the bar if the bar slides at a velocity $\vec{V} = 10\hat{a}_y \text{ m/s}$ and $\vec{B} = 3\hat{a}_z \text{ mWb/m}^2$. [2+3]

8. A uniform plane wave in free space is given by $\vec{H}_S = (250 \angle 30^\circ) e^{-j350Z} \hat{a}_x$ V/m. Determine phase constant, frequency of the wave, intrinsic impedance, $\vec{E}_S$ and the magnitude $\vec{H}$ of at $z = 25\text{mm}$ and $t = 4\text{ps}$. [1+2+1+2+2]
9. Within a certain region, $\epsilon = 10^{-11}\text{F/m}$ and $\mu = 10^{-5}\text{H/m}$. If $B_x = 2 \times 10^{-4} \cos 10^5 t \sin 10^{-3} y$ T find: [3+3+2]
- Find $\vec{E}$
 - Find the total magnetic flux passing through the surface $x = 0$, $0 < y < 40\text{m}$, $0 < z < 2\text{m}$ at $t = 1 \mu\text{s}$
 - Find the value of the closed line integral of $\vec{E}$ around the perimeter of the given surface.
10. A transmission line operating at 120MHz has $R = 20\Omega/\text{m}$, $L = 0.3\mu\text{H/m}$, $C = 63\text{pF/m}$ and $G = 4.2\text{ms/m}$. Find [3+3+2]
- Propagation coefficient (γ)
 - Velocity of wave propagation on the line (v)
 - Characteristic impedance (Z_0)
11. A rectangular waveguide has dimension $a = 4\text{cm}$ and $b = 2\text{cm}$. Determine the cut-off frequency and range of frequencies over with the guide will operate single mode. [6]
12. Write short notes on antenna and its types. [2]

TRIBHUVAN UNIVERSITY
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2076 Ashwin

Exam.	Back		
	Level	Full Marks	80
Programme	BEL, BEX, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electromagnetics (EX 503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary charts and codes are attached herewith.
- ✓ Assume suitable data if necessary.

1. Given points A ($\rho = 5$, $\phi = 70^\circ$, $z = -3$) and B ($\rho = 2$, $\phi = -30^\circ$, $z = 1$), find: (a) a unit vector in cartesian coordinates at A directed toward B; (b) a unit vector in cylindrical coordinates at A directed toward B. [5]
2. Two uniform line charges, each 20 nC/m, are located at $y = 1$, $z = \pm 1$ m. Find the total electric flux leaving the surface of a sphere having a radius of 2 m, if it is centered at A (3, 1, 0). [6]
3. -Derive Energy Density in electrostatic field. [7]
4. The conducting planes $2x + 3y = 12$ and $2x + 3y = 18$ are at potentials of 100 V and 0, respectively. Let $\epsilon = \epsilon_0$ and find: a) V at P (5, 2, 6); b) E at P(5, 2, 6). [7]
5. Let a filamentary current of 5 mA be directed from infinity to the origin on the positive z axis and then back out to infinity on the positive x axis. Find H at P (0, 1, 0). [8]
6. State Ampere's circuital law. Let the permittivity be 5 $\mu\text{H/m}$ in region A where $x < 0$, and 20 $\mu\text{H/m}$ in region B where $x > 0$. If there is a surface current density $\mathbf{K} = 150\mathbf{a}_y - 200\mathbf{a}_z$ A/m at $x = 0$, and if $\mathbf{H}_A = 300\mathbf{a}_x - 400\mathbf{a}_y + 500\mathbf{a}_z$ A/m, find: (a) $|\mathbf{H}_{LA}|$; (b) $|\mathbf{H}_{NA}|$; (c) $|\mathbf{H}_{LB}|$; (d) $|\mathbf{H}_{NB}|$. [10]
7. State and explain the Maxwell's equation in differential and integral form. Also define the displacement current and depth of penetration. [10]
8. Establish the relation for Helmholtz's equation for electromagnetic wave propagation. [5]
9. State and prove Poynting's theorem. [6]
10. A load $Z_L = 80 + j100\Omega$ is located at $z = 0$ on a lossless 50- Ω line. The operating frequency is 200 MHz and the wavelength on the line is 2 m. (a) If the line is 0.8 m in length, use the Smith chart to find the input impedance. (b) What is S? (c) What is the distance from the load to the nearest voltage maximum? [7]
11. An air-filled rectangular waveguide has dimensions $a = 2$ cm and $b = 1$ cm. Determine the range of frequencies over which the guide will operate single mode (TE_{10}). [3]
12. Write short notes on: [3x2]
 - a) TE mode and TM mode
 - b) Antenna Properties

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
Examination Control Division
2075 Chaitra

Exam.	Regular / Back		
Level	BE	Full Marks	80
Programme	BEL, BEX, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electromagnetics (EX 503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary figures are attached herewith.
- ✓ Assume suitable data if necessary.
- ✓ Assume that the **Bold Faced** letter represents a vector and $\mathbf{a}_{\text{subscript}}$ represents a unit vector.

- Find the vector that extends from A(-3,-4,6) to B(-5,2,-8) and express it in cylindrical coordinate system. [1+4]
- A point charge of 12nC is located at the origin. Four uniform line charges are located in the $x=0$ plane as follow: 80nC/m at $y=-1$ and -5 m, -50 nC/m at $y=-2$ and -4 m. Find the electric flux density $\mathbf{D}$ at P(0,-3,2). [7]
- Let the region $z<0$ be composed of a uniform dielectric material for which $\epsilon_{R1}=3.2$, while the region $z>0$ is characterized by $\epsilon_{R2}=2$. Let $\mathbf{D}_1=-30\mathbf{a}_x+50\mathbf{a}_y+70\mathbf{a}_z$ nC/m² and find:- [7]
 - $\mathbf{D}_{t1}$ (Tangential component of $\mathbf{D}$ in Region 1);
 - Polarization ($\mathbf{P}_1$);
 - $\mathbf{E}_{n2}$ (Normal component of $\mathbf{E}$ in Region 2)
 - $\mathbf{E}_{t2}$ (Tangential component of $\mathbf{E}$ in Region 2)
- Derive the Poisson's and Laplace's equations. Assuming that the potential V in the cylindrical coordinate system is the function of 'r' only, solve the Laplace's equation by Integration Method and derive the expression for the capacitance of the Spherical Capacitor using the same solution of V . [2+5]
- Derive the equation for magnetic field intensity in different regions due to a co-axial cable carrying a uniformly distributed dc current I in the inner conductor and $-I$ in the outer conductor. [6]
- Find the vector magnetic field intensity $\mathbf{H}$ in Cartesian coordinate at P(-1.5, -4, 3) caused by a current filament of 12A in the $\mathbf{a}_z$ direction on the z -axis and extending from $z=-3$ to $z=3$. [6]
- Define Curl and give the physical interpretation of the Curl with a suitable example. [1+3]
- A uniform plane wave in free space is propagating in the $-\mathbf{a}_y$ direction at a frequency of 5 MHz. If $\mathbf{E}=200 \cos(\omega t + \beta y) \mathbf{a}_z$ V/m, write the expressions for electric and magnetic fields, i.e., $\mathbf{E}_s(x,y,z)$ and $\mathbf{H}_s(x,y,z)$ respectively in phasor forms. [3+5]
- Derive an expression for Standing Wave Ratio (SWR) indicating where on the z -axis you'll get the maximum and minimum value of electric field intensity E . Assume that the boundary is at $z=0$, the region $z<0$ is a perfect dielectric and the region $z>0$ may be of any material. [8]

10. Find the amplitude of the displacement current density in an air space within a large power transformer where $\mathbf{H} = 10^6 \cos(377t + 1.2566 \times 10^{-6}z) \mathbf{a}_y$ A/m. [6]
11. A lossless $50\text{-}\Omega$ line is 1.5λ long and is terminated with a pure resistance of 100Ω . The load voltage is $40/60^\circ$ V. Find: (a) the average power delivered to the load; (b) the magnitude of the minimum voltage on the line. [4+4]
12. What are the advantages and disadvantages of waveguides when you compare it with transmission lines? Explain the transverse electric (TE) and transverse magnetic (TM) modes used in rectangular waveguides. [3+3]
13. Give the definition of an antenna and explain the properties of any one type of antenna that you have studied during your electromagnetics course. [1+1]

Exam.	Back		
Level	BE	Full Marks	80
Programme	BEL, BEX, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electromagnetics (EX503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ The $\hat{a}_{\text{subscript}}$ denotes a unit vector along the direction of subscript.
- ✓ Necessary formulas are attached herewith.
- ✓ Assume suitable data if necessary.

1. Express in cartesian components: (a) the vector at A($\rho = 4$, $\Phi = 40^\circ$, $z = -2$) that extends to B($\rho = 5$, $\Phi = -110^\circ$, $z = 2$); (b) a unit vector at B directed toward A. [3+2]
2. Derive an Electric Field Intensity ($\vec{E}$) in between the two co-axial cylindrical conductors, the inner of radius 'a' and outer of radius 'b', each infinite in extent and assuming a surface charge density ρ_s on the outer surface of the inner conductor. An infinite uniform line charge $\rho_L = 2$ nC/m lies along the x-axis in free space, while the point charge of 8nC each are located at (0, 0, 1). Find $\vec{E}$ at (2, 3, -4) [4+4]
3. Derive the integral and point forms of continuity equation. In a certain region, $\vec{j} = 3r^2 \cos\theta \hat{a}_r - r^2 \sin\theta \hat{a}_\theta$ A/m². Find the current crossing the surface defined by $\theta = 30^\circ$, $0 < \phi < 2\pi$, $0 < r < 2$. [5+3]
4. Given the field, $\vec{D} = \frac{5 \sin(\theta) \cos(\phi)}{r} \hat{a}_r$ C/m², find: (a) the volume charge density; (b) the total charge contained in the region $r < 2$ m; (c) the value of D at the surface $r = 2$. [2+2+2]
5. Differentiate between scalar and vector magnetic potential. Derive the expression for magnetic boundary conditions. [3+5]
6. State Stroke's theorem. Evaluate both sides of Stroke's theorem for the field $\vec{G} = 10 \sin\theta \hat{a}_\phi$ and the surface $r = 3$, $0 \leq \theta \leq 2\pi$, $0 \leq \phi \leq 90^\circ$. Let the surface have the $\hat{a}_r$ direction. [1+7]
7. Find the capacitance of a spherical capacitor using Laplace's equation. [6]
8. Write point form of all the Maxwell's Equations in phasor domain, for perfect dielectric material. Use these equations to derive the magnetic field component of a uniform plane wave travelling in the perfect dielectric medium. [2+6]
9. Let $\vec{E}(z, t) = 1800 \cos(10^7 \pi t - \beta z) \hat{a}_x$ V/m and $\vec{H}(z, t) = 3.8 \cos(10^7 \pi t - \beta z) \hat{a}_y$ A/m represents a uniform plane wave propagating at a velocity of 1.4×10^8 m/s in perfect dielectric. Find a) β b) λ c) η d) μ_r e) ϵ_r . [2+1+2+2+1]

10. The velocity of propagation in a lossless transmission line 2.5×10^8 m/s. If the capacitance of the line is 30 pf/m, find: [2+2+2+2]
- a) Inductance of the line
 - b) Characteristic impedance
 - c) Phase constant at 100 MHz
 - d) Reflection coefficient if the line is terminated with a resistive load of 50Ω
11. What are the advantages of waveguides over transmission lines? A rectangular waveguide has a cross-section of $2.5 \text{ cm} \times 1.2 \text{ cm}$. Find the cut-off frequencies at dominant mode and TE (1,1) [1+4]
12. Write short notes on: Antenna properties [2]

Exam.	Regular		
	BE	Full Marks	80
Level	BEL, BEX, BCT	Pass Marks	32
Programme	II / I	Time	3 hrs.

Subject: - Electromagnetics (EX503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ The $\vec{a}_{\text{subscript}}$ denotes a unit vector along the direction of subscript.
- ✓ Necessary formulas are attached herewith.
- ✓ Assume suitable data if necessary.

1. An uniform Electric Field Intensity in certain region is given by $\vec{E} = y\vec{a}_x - xy\vec{a}_y + z\vec{a}_z$. Transform this field vector into cylindrical co-ordinate at a point P(2, 45°, 3). [5]
2. A uniform line charge density of 150 $\mu\text{C/m}$ lies at $x = 2$, $z = -4$ and a uniform sheet of charge equal to 25 nC/m^2 is placed at $z = 5$ plane. Find $\vec{D}$ at point (1, 2, 4) and convert it to the spherical coordinate system. [5+3]
3. Given the potential function $V = \frac{20 \cos \theta}{r^2} \text{ V}$ in free space and point P is located at $r = 3\text{m}$, $\theta = 60^\circ$, $\phi = 30^\circ$ find: a) $\vec{E}_P$ b) $\frac{dV}{dN}$ at P c) unit normal vector at P d) ρ_v at P. [2+1+1+2]
4. Define Relaxation time Constant (RTC). Derive an expression for RTC. Given the vector current density $\vec{J} = 10\rho^2 z \vec{a}_\rho - 4\rho \cos^2 \phi \vec{a}_\phi \text{ mA/m}^2$. Find the current flowing outward through the circular band $\rho = 3$, $0 < \phi < 2\pi$, $2 < z < 2.8$. [1+3+4]
5. Show that the vector magnetic potential can be defined in both the regions where $\vec{J}$ is equal or non-equal to zero. Use the concept of vector magnetic potential to derive the Magnetic Field Intensity due to an infinite current carrying filament carrying DC current I. [3+5]
6. State Stoke's theorem. Given the field $\vec{H} = \frac{1}{2} \cos\left(\frac{\phi}{2}\right) \vec{a}_\rho - \sin\left(\frac{\phi}{2}\right) \vec{a}_\phi \text{ A/m}$, evaluate both sides of Stoke's theorem for the path formed by the intersection of the cylinder $\rho = 3$ and the plane $z = 2$, and for the surface defined by $\rho = 3$, $0 \leq z \leq 2$, and $z = 0$, $0 \leq \rho \leq 3$. [1+7]
7. State Faradays Law. Correct the equation $\nabla \times \vec{H} = \vec{J}$ with necessary arguments and derivation for time varying field. [2+4]
8. Derive the expressions for reflection coefficient and transmission coefficient for the reflection of uniform waves at normal incidence. [8]

9. At 50 MHz, a lossy dielectric material is characterized by $\epsilon = 3.6\epsilon_0$, $\mu = 2.1\mu_0$ and $\sigma = 0.08 \text{ S/m}$. If $\vec{E}_s = 6e^{-\gamma x} \vec{a}_z \text{ V/m}$, Compute:

[2+2+4]

- Propagation Constant
- Wavelength
- $\vec{H}_s$

10. State the condition for lossless transmission line. A lossless transmission line is 80 cm long and operates at a frequency of 600 MHz. The line parameters are $L = 0.25 \mu\text{H/m}$ and $C = 100 \text{ pF/m}$. Find a) characteristics impedance b) phase constant c) phase velocity. [1+2+3+2]

11. Differentiate between Transmission line and waveguide. Consider a rectangular waveguide with $\epsilon_r = 2$, $\mu_r = 1$ with dimensions $a = 1.07 \text{ cm}$, $b = 0.43 \text{ cm}$ find the cut off frequency for TM_{11} mode and the dominant mode.

[1+4]

12. Write short notes on antenna and its parameters.

[2]

P.T.O

Exam.	Back		
Level	BE	Full Marks	80
Programme	BEL, BEX, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electromagnetics (EX503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt *all* questions.
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- ✓ Assume suitable data if necessary.

- Convert the vector $\vec{F} = F_x \vec{a}_x + F_y \vec{a}_y + F_z \vec{a}_z$ to both spherical coordinate system. [5]
- Find the electric field intensity in all three regions due to an infinite sheet parallel plate capacitor having surface charge density ρ_s C/m² and $-\rho_s$ C/m² and placed at $y = 0$ and $y = b$ respectively. Let a uniform line charge density, 3 nC/m, at $y = 3$; uniform surface charge density, 0.2 nC/m² at $x = 2$. Find $\vec{E}$ at the origin. [4+4]
- What is dipole? Derive the equation for potential and electric field due to dipole at a distant point P. [1+6]
- Derive Poisson's equation. By solving Laplace's equation, find the capacitance of a parallel plate capacitor with potential difference between the plates equals V_0 . [1+5]
- Verify stoke's theorem for the field $\vec{H} = \left(\frac{3r^2}{\sin \theta} \right) \vec{a}_\theta + 54r \cos \theta \vec{a}_\phi$ A/m in free space for the conical surface defined by $\theta = 20^\circ$, $0 \leq \phi \leq 2\pi$, $0 \leq r \leq 5$. Let the positive direction of $d\vec{s}$ be $\vec{a}_\theta$. [8]
- Consider a boundary at $z = 0$ for which $\vec{B}_1 = 2\vec{a}_x - 3\vec{a}_y + \vec{a}_z$ mT, $\mu_1 = 4$ μ H/m ($z > 0$), $\mu_2 = 7$ μ H/m ($z < 0$) and $\vec{K} = 80\vec{a}_x$ A/m at $z = 0$. Find $\vec{B}_2$ [8]
- Explain how Ampere's law conflict with continuity equation and how it is corrected? Derive conduction and displacement current in a capacitor. [4+3]
- Derive the expression for electric and magnetic fields for a uniform plane wave propagating in a perfect dielectric medium. [5+3]
- A 9.4 GHz uniform plane wave is propagating in a medium with $\epsilon_r = 2.25$ and $\mu_r = 1$. If the magnetic field intensity is 7 mA/m and the material is loss less, find [1+1+1+2+2]
 - Velocity of propagation
 - The wave length
 - Phase constant
 - Intrinsic impedance
 - Magnitude of electric field intensity

10. A lossless line having an air dielectric has a characteristics impedance of 400Ω . The line is operating at 200 MHz and $z_{in} = 200 - j200 \Omega$. Find (a) SWR (b) Z_L , if the line is 1 m long; (c) the distance from the load to the nearest voltage maximum. [2+4+2]
11. Differentiate between transmission line and waveguide. A rectangular waveguide having cross-section of $2 \text{ cm} \times 1 \text{ cm}$ is filled with a lossless medium characterized by $\epsilon = 4\epsilon_0$ and $\mu_r = 1$. Calculate the cut-off frequency of the dominant mode. [4+2]
12. Write short notes on antenna and its properties. [2]

DIVERGENCE

$$\text{CARTESIAN} \quad \nabla \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\text{CYLINDRICAL} \quad \nabla \cdot \vec{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

$$\text{SPHERICAL} \quad \nabla \cdot \vec{D} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin \theta} \frac{\partial (\sin \theta D_\theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi}$$

GRADIENT

$$\text{CARTESIAN} \quad \nabla V = \frac{\partial V}{\partial x} \hat{a}_x + \frac{\partial V}{\partial y} \hat{a}_y + \frac{\partial V}{\partial z} \hat{a}_z$$

$$\text{CYLINDRICAL} \quad \nabla V = \frac{\partial V}{\partial \rho} \hat{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \hat{a}_\phi + \frac{\partial V}{\partial z} \hat{a}_z$$

$$\text{SPHERICAL} \quad \nabla V = \frac{\partial V}{\partial r} \hat{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{a}_\phi$$

CURL

$$\text{CARTESIAN} \quad \nabla \times \vec{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \hat{a}_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \hat{a}_y + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \hat{a}_z$$

$$\text{CYLINDRICAL} \quad \nabla \times \vec{H} = \left(\frac{1}{\rho} \frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z} \right) \hat{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho} \right) \hat{a}_\phi + \frac{1}{\rho} \left(\frac{\partial (\rho H_\phi)}{\partial \rho} - \frac{\partial H_\rho}{\partial \phi} \right) \hat{a}_z$$

$$\text{SPHERICAL} \quad \nabla \times \vec{H} = \frac{1}{r \sin \theta} \left(\frac{\partial (H_\phi \sin \theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi} \right) \hat{a}_r + \frac{1}{r} \left(\frac{\partial H_r}{\partial \sin \theta} - \frac{\partial (r H_\phi)}{\partial r} \right) \hat{a}_\theta + \frac{1}{r} \left(\frac{\partial (r H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta} \right) \hat{a}_\phi$$

LAPLACIAN

$$\text{CARTESIAN} \quad \nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\text{CYLINDRICAL} \quad \nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\text{SPHERICAL} \quad \nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

Exam.	Regular		
Level	BE	Full Marks	80
Programme	BEL, BEX, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electromagnetics (EX503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
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- ✓ Assume suitable data if necessary.

- Express a scalar potential field $V = x^2 + 2y^2 + 3z^2$ in spherical coordinates. Find value of V at a point $P(2, 60^\circ, 90^\circ)$. [3+2]
- Derive the expression of Electric field intensity due to a line charge using Gauss Law. Find Electric flux density at point $P(5, 4, 3)$ due to a uniform line charge of 2 nC/m at $x = 5$, $y = 3$, point charge 12 nC at $Q(2, 0, 6)$ and uniform surface charge density of 0.2 nC/m^2 at $x = 2$. [4+4]
- State the physical significance of divergence. Derive the Divergence theorem. Given the potential $V = \frac{10}{r^2} \sin\theta \cos\phi$; find the electric flux density $\vec{D}$ at $\left(2, \frac{\pi}{2}, 0\right)$. [2+2+3]
- Derive Laplace's equation. Find the capacitance of a co-axial cable using Laplace's equation. [1+5]
- State Ampere's circuital law. By using Biot Savart's law, derive an expression for magnetic field intensity $\left(\vec{H}\right)$ due to an infinite length filament carrying a direct current I . [2+6]
- Flux density at medium with $\mu_1 = 15$ is $\vec{B}_1 = 1.2\vec{a}_x + 8\vec{a}_y + 4\vec{a}_z \text{ T}$. Find $\vec{B}, \vec{H}$ and the angles between the field vectors and tangent to the interface at second medium, if second medium has $\mu_2 = 1$, and interface plane is $z = 0$. [3+2+3]
- State and derive the expression of motional emf (electromotive force). Consider two parallel conductors placed at $x = 0$ and $x = 5 \text{ cm}$ in a magnetic field $\vec{B} = 6\vec{a}_z \text{ mWb/m}^2$. A high resistance voltmeter is connected at one end and a conducting bar is sliding at other end with velocity $\vec{v} = 18\vec{a}_y \text{ m/s}$. Calculate the induced voltage and show the polarity of induced voltage across the voltmeter. [1+3+3]
- What is standing wave? Derive the equation of Electric field and Magnetic field and SWR of standing wave? [1+7]

9. An EM wave travels in free space with the electric field component $\vec{E} = (15\vec{a}_y - 5\vec{a}_z) \cos(\omega t - 3y + 5z) \text{ V/m}$. Find (a) ω and λ (b) the magnetic field component. [2+2+3]
10. A 50Ω lossless transmission line is 30 m long and is terminated with a load $Z_L = 60 + j40\Omega$. The operating frequency is 20 MHz and velocity on the line is $2.5 \times 10^8 \text{ m/s}$. Find [2+2+4]
- Reflection coefficient
 - Standing wave ratio
 - Input impedance
11. Explain TE and TM modes? Consider a rectangular waveguide with $\epsilon_r = 2.25$ and $\mu_r = 1$ with dimensions $a = 1.07$, $b = 0.43$. Find the cut-off frequency for TM_{11} mode and dominant mode. [2+4]
12. Write short notes on antenna and its type. [2]

New Back (2066 & Later Batch)			
Exam.	BE	Full Marks	80
Level	BEL, BEX, BCT	Pass Marks	32
Programme	II/I	Time	3 hrs.

Subject: - Electromagnetics (EX503)

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- ✓ Attempt **All** questions.
- ✓ The figures in the margin indicate **Full Marks**.
- ✓ **Necessary tables are attached herewith.**
- ✓ $\vec{A}$ represent a vector and $\hat{a}_{\text{subscript}}$ and $\vec{a}_{\text{subscript}}$ denotes a **unit** vector along the direction given by the subscript.
- ✓ Assume suitable data if necessary.

1. Define a vector field. A field vector is given by an expression

$$\vec{A} = \frac{1}{\sqrt{x^2 + y^2 + z^2}} (x\vec{a}_x + y\vec{a}_y + z\vec{a}_z), \text{ transform this vector in cylindrical coordinate system at point } (2, 30^\circ, 6) \quad [2+3]$$

2. Given the flux density $\vec{D} = (2 \cos \theta / r^3) \vec{a}_r + (\sin \theta / r^3) \vec{a}_\theta$ C/m², evaluate both sides of the divergence theorem for the region defined by $1 < r < 2, 0 < \theta < \frac{\pi}{2}, 0 < \phi < \frac{\pi}{2}$. [8]

3. Define electric dipole and polarization. The region $z < 0$ contains a dielectric material for which $\epsilon_{r1} = 2.5$ while the region $z > 0$ is characterized by $\epsilon_{r2} = 4$. Let $\vec{E}_1 = -30\hat{a}_x + 50\hat{a}_y + 70\hat{a}_z$ V/m. Find: (a) $\vec{E}_2$ (b) $\vec{D}_2$ (c) polarization in region 2 $\left(\vec{P}_2 \right)$. [2+2+2+1+1]

4. State the uniqueness theorem and prove this theorem for Laplace's equation. [1+5]

5. A current density in certain region is given as: $\vec{J} = 20 \sin \theta \cos \phi \vec{a}_r + \frac{1}{r} \vec{a}_\phi$ A/m², Find: [5+3]

i) The average value of J_r over the surface $r = 1, 0 < \theta < \pi/2, 0 < \phi < \pi/2$

ii) $\frac{\partial \rho_v}{\partial t}$

6. Show that $\nabla \times \vec{E} = 0$ for static electric field. The region $y < 0$ (Region 1) is air and $y > 0$ (Region 2) has $\mu_r = 10$. If there is a uniform magnetic field $\vec{H} = 5\vec{a}_x + 6\vec{a}_y + 7\vec{a}_z$ A/m in region 1, find $\vec{B}$ and $\vec{H}$ in region 2. [2+3+3]

7. Find the amplitude of the displacement current density in a metallic conductor at 60 Hz, if $\epsilon = \epsilon_0, \mu = \mu_0, \sigma = 5.8 \times 10^7$ S/m, and $\vec{J} = \sin(377t - 117.1z) \vec{a}_x$ MA/m². [5]

8. Explain the phenomena when a plane wave is incident normally on the interface between two different Medias. Derive the expression for reflection and transmission coefficient. [8]
9. A uniform plane wave in non-magnetic medium has $\vec{E} = 50 \cos(10^8 t + 2z) \hat{a}_y$ V/m . Find:
 i) The direction of propagation
 ii) Phase constant β , wavelength λ , velocity v_p , relative permittivity ϵ_r , intrinsic impedance η
 iii) $\vec{H}$ [1+5+2]
10. Determine the primary constants (R, L, C and G) on the transmission line when the measurement on the line at 1 KHz gave the following results: $z_0 = 710 \angle -16^\circ$, $\alpha = 0.01$ neper/m and $\beta = 0.035$ rad/m. [8]
11. Explain the modes supported by a rectangular waveguide. Calculate the cut off frequencies of the first four propagating modes for an air filled copper waveguide with dimension $a = 2.5$ cm, $b = 1.2$ cm. [2+4]
12. Write short notes on antenna and its types. [2]

Exam.	Regular		
	Level	Full Marks	80
Programme	BEL, BEX, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electromagnetics (EX503)

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✓ Attempt All questions.

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✓ Necessary tables are attached herewith.

✓ $\vec{A}$ represent a vector and $\hat{a}_{\text{subscript}}$ and $\vec{a}_{\text{subscript}}$ denotes a unit vector along the direction given by the subscript.

✓ Assume suitable data if necessary.

- Express the uniform vector field $\vec{F} = 5\vec{a}_x$ in (a) cylindrical components (b) spherical components. [2+3]
- Derive the expression for the electric field intensity due to an infinitely long line charge with uniform charge density ρ_L by using Gauss's law. A uniform line charge density of 20 nC/m is located at $y=3$ and $z=5$. Find $\vec{E}$ at $P(5,6,1)$ [4+4]
- Derive an expression to calculate the potential due to a dipole in terms of the dipole moment $\left(\vec{p}\right)$. A dipole for which $\vec{p} = 3\vec{a}_x - 5\vec{a}_y + 10\vec{a}_z$ nC.m is located at the point $(1,2,-4)$. Find $\vec{E}$ at P . [4+4]
- Assuming that the potential V in the cylindrical coordinate system is function of ρ only, solve the Laplace's equation and derive the expression for the capacitance of coaxial capacitor of length L using the same solution of V . Assume the inner conductor of radius a is at potential V_0 with respect to the conductor of radius b . [6]
- State and derive expression for Stoke's theorem. Evaluate the closed line integral of $\vec{H}$ from $P_1(5,4,1)$ to $P_2(5,6,1)$ to $P_3(0,6,1)$ to $P_4(0,4,1)$ to P_1 using straight line segments, if $\vec{H} = 0.1y^3 \vec{a}_x + 0.4xz \vec{a}_z$ A/m. [1+3+4]
- Define scalar magnetic potential and show that it satisfies the Laplace's equation. Given the vector magnetic potential $\vec{A} = -(\rho^2/4)\vec{a}_z$ Wb/m, calculate the total magnetic flux crossing the surface $\phi = \pi/2$, $1 \leq \rho \leq 2$ m and $0 \leq z \leq 5$ m. [1+2+5]
- How does $\nabla \times \vec{H} = \vec{J}$ conflict with continuity equation in time varying fields. How is this conflict rectified in such fields? [2+3]
- Derive the expression for electric and magnetic fields for a uniform plane wave propagating in a perfect dielectric space. [5+3]
- A lossless dielectric material has $\sigma=0, \mu_r=1, \epsilon_r=4$. An electromagnetic wave has magnetic field expressed as $\vec{H} = -0.1 \cos(\omega t - z)\vec{a}_x + 0.5 \sin \cos(\omega t - z)\vec{a}_y$ A/m. Find: [2+2+4]
 - Angular frequency (ω)
 - Wave impedance (η)
 - $\vec{E}$

10. Consider a two-wire 40Ω line ($Z_0 = 40 \Omega$) connecting the source of 80 V, 400 kHz with series resistance 10Ω to the load of $Z_L = 60 \Omega$. The line is 75 m long and the velocity on the line is 2.5×10^8 m/s. Find the voltage $V_{in,s}$ at input end and $V_{L,s}$ at output end of the transmission line.

[8]

11. Why does a hollow rectangular waveguide not support TEM mode? A rectangular air-filled waveguide has a cross-section of 45×90 mm. Find the cut-off frequencies of the first four propagating modes.

[2+4]

12. Write short notes on antenna and its types.

[2]

DIVERGENCE

CARTESIAN $\nabla \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$

CYLINDRICAL $\nabla \cdot \vec{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$

SPHERICAL $\nabla \cdot \vec{D} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin \theta} \frac{\partial (\sin \theta D_\theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi}$

GRADIENT

CARTESIAN $\nabla V = \frac{\partial V}{\partial x} \hat{a}_x + \frac{\partial V}{\partial y} \hat{a}_y + \frac{\partial V}{\partial z} \hat{a}_z$

CYLINDRICAL $\nabla V = \frac{\partial V}{\partial \rho} \hat{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \hat{a}_\phi + \frac{\partial V}{\partial z} \hat{a}_z$

SPHERICAL $\nabla V = \frac{\partial V}{\partial r} \hat{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{a}_\phi$

CURL

CARTESIAN $\nabla \times \vec{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \hat{a}_x + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \hat{a}_y + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \hat{a}_z$

CYLINDRICAL $\nabla \times \vec{H} = \left(\frac{1}{\rho} \frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z} \right) \hat{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho} \right) \hat{a}_\phi + \frac{1}{\rho} \left(\frac{\partial (\rho H_\phi)}{\partial \rho} - \frac{\partial H_\rho}{\partial \phi} \right) \hat{a}_z$

SPHERICAL $\nabla \times \vec{H} = \frac{1}{r \sin \theta} \left(\frac{\partial (H_\phi \sin \theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi} \right) \hat{a}_r + \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial H_r}{\partial \phi} - \frac{\partial (r H_\phi)}{\partial r} \right) \hat{a}_\theta + \frac{1}{r} \left(\frac{\partial (r H_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta} \right) \hat{a}_\phi$

LAPLACIAN

CARTESIAN $\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$

CYLINDRICAL $\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$

SPHERICAL $\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$

Exam.	New Back (2066 & Later Batch)		
Level	BE	Full Marks	80
Programme	BEL, BEX, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electromagnetics (EX503)

- ✓ Candidates are required to give their answers in their own words as far as practicable.
- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formula sheet is attached herewith.
- ✓ $\vec{a}_{\text{subscript}}$ denote a unit vector along the direction given by the subscript.
- ✓ Assume suitable data if necessary.

1. Transform $\vec{A} = 10\vec{a}_x - 8\vec{a}_y + 6\vec{a}_z$; at point p(10,-8,6) to cylindrical coordinate system. [5]
2. A line charge of 8nC/m is located at $x = -1$, $y = 2$, a point charge of 6mC at $y = -4$ and a surface charge of 30 pC/m² at $z = 0$. If the potential at origin is 100V, find the potential at P (4,1,3). [7]
3. Explain the Continuity equation. The current density in certain region is approximated by $\vec{J} = \left(\frac{0.1}{r}\right)e^{-10^6 t} \vec{a}_r$ A/m² in spherical coordinates. (a) How much current is crossing the surface $r = 50\text{cm}$ at $t = 1\mu\text{S}$? (b) Find $\rho_v(r,t)$ assuming that $\rho_v \rightarrow 0$ as $t \rightarrow \infty$. [2+6]
4. Find the equation for Energy Density in the electrostatic field. [6]
5. Differentiate between scalar and vector magnetic potential. Derive an expression for the magnetic field intensity $(\vec{H})$ at a point due to an infinite filament carrying a direct current I, placed on z-axis using ampere's circuital law. [2+6]
6. State and prove Stoke's theorem. Given $\vec{H} = 10\sin\theta\vec{a}_r$ in free space. Find the current in $\vec{a}_r$ direction having $r = 3, 0 \leq \theta \leq 90^\circ, 0 \leq \phi \leq 90^\circ$. [3+5]
7. Within a certain region, $\epsilon = 10^{-11}$ F/m and $\mu = 10^{-5}$ H/m.
If $\vec{B}_x = 2 \times 10^{-4} \cos 10^5 t \sin 10^{-3} y \vec{a}_x$ T: (a) Use $\nabla \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t}$ to find $\vec{E}$; (b) Find the total magnetic flux passing through the surface $x = 0, 0 \leq y \leq 40\text{m}, 0 \leq z \leq 2\text{m}$, at $t = 1\mu\text{S}$. [4+4]
8. Derive an expression for standing wave ratio of uniform plane wave in terms of reflection coefficient. Find the reflection coefficient for the interface between air and fresh water ($\epsilon = 81\epsilon_0, \sigma \cong 0$), in case of normal incidence. [5+3]

9. The magnetic field intensity $\left(\vec{H}\right)$ in free space is given as,

$$\vec{H}(x,t) = 10 \cos(10^8 t + \beta x) \vec{a}_y \text{ A/m find:}$$

[2+1+3]

- a) Phase constant (β)
- b) Wavelength
- c) $\left|\vec{E}(x,t)\right|$ at P (0.1, 0.2, 0.3) at $t = 1 \text{ nS}$

10. A 300Ω transmission line is lossless, 0.25λ long, and is terminated in $Z_L = 500 \Omega$. The line has a generator with $90 \angle 0^\circ \text{ V}$ in series with 100Ω connected to the input. Find (a) the load voltage (b) voltage at the midpoint of the line.

[4+4]

11. Determine the cut-off frequency for an air filled rectangular waveguide with $a = 2.5 \text{ cm}$ and $b = 1.25 \text{ cm}$ for TE_{11} mode.

[4]

12. Write short notes on:

[2+2]

- a) Loss tangent
- b) Antenna types and properties

Exam.	Regular		
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Programme	BEL, BEX, BCT	Pass Marks	32
Year / Part	II / I	Time	3 hrs.

Subject: - Electromagnetic (EX503)

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- ✓ Attempt All questions.
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1. Transform the vector $4\vec{a}_x - 2\vec{a}_y - 4\vec{a}_z$ into spherical coordinates at point P(-2,-3,4) [5]
2. State and write the mathematical equation of Gauss Law. Using the same law, derive an expression for electric field intensity ($\vec{E}$) in between the two co-axial cylindrical conductors having inner radius 'a' and outer radius 'b', each infinite in extent and assuming a surface charge density ρ_s on the outer surface of the inner conductor. [1+6]
3. State the physical significance of potential gradient. Assuming that the potential V in the spherical coordinate system is a function of r only, solve the Laplacian equation and derive the expression for the capacitance of a spherical capacitor using the same solution of V. [2+6]
4. Within the cylinder $\rho = 2$, $0 < z < 1$ the potential given by: $V = 100 + 50\rho + 150\rho\sin\phi$ V find: [2+1+2+1]
 - a) Electric Field Intensity ($\vec{E}$) at P (1, 60° , 0.5) in free space
 - b) Potential Gradient $\left(\frac{dV}{dN}\right)$
 - c) Volume Charge Density (ρ_v) at P(1, 60° , 0.5) in free space
 - d) How much charge lies within the cylinder?
5. State the physical significance of Curl. Evaluate both sides of stokes theorem for the field $\vec{A} = 6xy \vec{a}_x - 3y^2 \vec{a}_y$ A/m and the rectangular path around the region; $2 \leq x \leq 5$, $-1 \leq y \leq 1$, $z = 0$. Let the positive direction of $\vec{ds}$ be $\vec{a}_z$. [2+6]
6. Explain the physical significance of the equation $\oint \vec{B} \cdot d\vec{s} = 0$. Given the vector magnetic potential $\vec{A} = \rho^2 / 8 \vec{a}_z$ Wb/m. Calculate the total magnetic flux crossing the surface $\phi = \pi/4$, $1 \leq \rho \leq 3$ m, $0 \leq z \leq 5$ m. [2+6]

7. Explain motional emf and transformer emf with necessary mathematical derivations. A straight conductor of 0.2m lies along x-axis with one end at the origin. If this conductor is subjected to the magnetic flux density $\vec{B} = 0.08\vec{a}_y$ T and velocity $\vec{v} = 2.5 \sin 10^3 t \vec{a}_z$ m/s. Calculate the emf induced in the conductor. [6+2]
8. Define Transverse Electromagnetic (TEM) wave. Derive an expression electric field for a uniform plane wave propagating in a perfect dielectric media. [7+1]
9. A uniform plane wave in free space at a frequency of 12 MHz is given by $\vec{E} = 200 \cos(\omega t + 120x + 30^\circ) \vec{a}_y$ V/m, find (a) $|E_{\max}|$ (b) $\vec{H}$ at $x = 40\text{mm}$ and $t = 340\text{ps}$. [3+3]
10. A lossless transmission line with $Z_0 = 50\Omega$ has a length of 0.4λ . The operating frequency is 300MHz and it is terminated with a load $Z_L = 40 + j30\Omega$. Find: [2+2+4]
 - a) Reflection coefficient (Γ)
 - b) Standing wave ratio on the line (SWR)
 - c) Input impedance (Z_{in})
11. Explain Transverse Electric Mode and Transverse Magnetic Mode of a waveguide. [2+2]
12. Write short notes on: [2×2]
 - a) Skin depth
 - b) Antenna and its types

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Subject: - Electromagnetics (EX503)

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- ✓ Attempt All questions.
- ✓ The figures in the margin indicate Full Marks.
- ✓ Necessary formulas are attached herewith.
- ✓ Assume suitable data if necessary.
- ✓ Assume that the Bold Faced letter represents a vector and $\mathbf{a}_{\text{subscript}}$ represents a unit vector.

- (1) Express the vector field, $\mathbf{G} = (x^2 + y^2)^{-1/2}(\mathbf{x}\mathbf{a}_x + \mathbf{y}\mathbf{a}_y)$ in cylindrical components and cylindrical variables. [5]
- (2) Find $\mathbf{D}$ at the point $(-3, 4, 2)$ if the following charge distributions are present in free space: point charge, $+2 \text{ nC}$, at $\mathbf{P}(2, 0, 6)$; uniform line charge density, 3 nC/m , at $x = -2, y = 3$; uniform surface charge density, 0.2 nC/m^2 at $x = 2$. [7]
- (3) Two uniform line charges, 8 nC/m each, are located at $x = 1, z = 2$, and at $x = -1, y = 2$ in free space. If the potential at the origin is 100V , find V at $\mathbf{P}(4, 1, 3)$. [7]
- (4) State the Uniqueness theorem and prove that the solution of Poisson's equation is unique. [1+6]
- (5) Write the equation of the Vector Magnetic Potential in differential form. Using the same equation, derive the equation for magnetic field intensity at a point due to an infinite filament carrying a uniformly distributed dc current I . [1+5]
- (6) Calculate the value of the vector current density: (a) in cylindrical coordinates at $\mathbf{P}_1 (\rho=1.5, \phi=90^\circ, z=0.5)$ if $\mathbf{H} = \frac{2}{\rho}(\cos 0.2\phi) \mathbf{a}_\phi$. [3+3]
(b) in spherical coordinates at $\mathbf{P}_2 (r=2, \theta=30^\circ, \phi=20^\circ)$ if $\mathbf{H} = \frac{1}{\sin\theta} \mathbf{a}_\theta$. [1+3]
- (7) State and derive the Stoke's theorem. [1+3]
- (8) What is an input intrinsic impedance? Derive an expression for the input intrinsic impedance using the concept of reflection of uniform plane waves. [2+6]

- (9) The electric field amplitude of a uniform plane wave propagating in the free space in $\mathbf{a}_z$ direction is 250 V/m. If $\mathbf{E} = E_x \mathbf{a}_x$ and $\omega = 1.00$ Mrad/s, find: (a) the frequency; (b) the wavelength; (c) the period; (d) the amplitude of $\mathbf{H}$. [2+2+1+3]
- (10) Find the amplitude of the displacement current density inside a typical metallic conductor where $f = 1$ kHz, Conductivity $\sigma = 5 \times 10^7$ mho/m, dielectric constant $\epsilon_R = 1$; and the conduction current density $\mathbf{J} = 10^7 \sin(6283 t - 444 z) \mathbf{a}_x$ A/m². [6]
- (11) A 50- Ω lossless line has a length of 0.4λ . The operating frequency is 300 MHz. A load $Z_L = 40 + j30 \Omega$ is connected at $z = 0$, and the Thevenin equivalent source at $z = -l$ is $12 \angle 0^\circ$ in series with $Z_{Th} = 50 + j0 \Omega$. Find: (a) The Reflection Coefficient Γ (b) The Voltage Standing Wave Ratio (VSWR) and (c) The input Impedance Z_{in} . [2+2+4]
- (12) Explain why is it not possible to use waveguides at lower frequencies? Explain the transverse electric (TE) and transverse magnetic (TM) modes used in rectangular waveguides. [2+4]
- (13) Give the definition of an antenna. Explain the properties of any one type of antenna that you have studied during your electromagnetics course. [1+1]

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1. Transform the Vector $\vec{A} = y\vec{a}_x + x\vec{a}_y + z\vec{a}_z$ into cylindrical co-ordinates at a point $p(2, 45^\circ, 5)$ [5]
2. Along the z-axis there is a uniform line of charge with $\rho_L = 4\pi \text{ Cm}^{-1}$ and in the $x = 1$ plane there is a surface charge with $\rho_s = 20 \text{ Cm}^{-2}$. Find the Electric Flux Density at $(0.5, 0, 0)$ [6]
3. Define Uniqueness theorem. Assuming that the potential V in the cylindrical coordinate system is the function of ' ρ ' only, solve the Laplacian Equation by integration method and derive the expression for the Capacitance of the co-axial capacitor using the same solution of V . [2+5]
4. Define Electric Dipole and Polarization. Consider the region $y < 0$ be composed of a uniform dielectric material for which the relative permittivity (ϵ_r) is 3.2 while the region $y > 0$ is characterized by $\epsilon_r = 2$. Let the flux density in region 1 be $\vec{D}_1 = -30\vec{a}_x + 50\vec{a}_y + 70\vec{a}_z \text{ nC/m}^2$. Find: [2+3+3]
 - a) Magnitude of Flux density and Electric fields intensity at region 2.
 - b) Polarization ($\vec{P}$) in region 1 and region 2
5. State Ampere's circuital law and stoke's theorem. Derive an expression for magnetic field intensity ($\vec{H}$) due to infinite current carrying filament using Biot Savart's Law. [1+2+5]
6. Differentiate between scalar and vector magnetic potential. The magnetic field intensity in a certain region of space is given as $\vec{H} = (2\rho + z)\vec{a}_\rho + \frac{2}{z}\vec{a}_z \text{ A/m}$. Find the total current passing through the surface $\rho = 2, \pi/4 < \phi < \pi/2, 3 < z < 5$, in the $\vec{a}_\rho$ direction. [3+5]
7. State Faraday's law and correct the equation $\nabla \times \vec{E} = 0$ for time varying field with necessary derivation. Also modify the equation $\nabla \times \vec{H} = \vec{J}$ with necessary derivations for time varying field. [1+3+4]
8. Derive an expression for input intrinsic Impedence using the concept of reflection of uniform plane waves. [6]

9. Find the amplitude of displacement current density inside a typical metallic conductor where $f = 1\text{KHz}$, $\sigma = 5 \times 10^7 \text{ mho/m}$, $\epsilon_r = 1$ and the conduction current density is $\vec{J} = 10^7 \sin(6283t - 444z) \hat{a}_y \text{ A/m}^2$ [4]
10. Write all the Maxwell equations for the time varying field point form as well as integral form. [4]
11. A lossless transmission line with $Z_0 = 50 \Omega$ with length 1.5 m connects a voltage $V_g = 60\text{V}$ source to a terminal load of $Z_L = (50 + j50) \Omega$. If the operating frequency $f = 100 \text{ MHz}$, generator impedance $Z_g = 50 \Omega$ and speed of wave equal to the speed of the light. Find the distance of the first voltage maximum from the load. What is the power delivered to the load? [4+4]
12. What are the techniques that can be taken to match the transmission line with mismatched load? Explain any one. [2]
13. Write short notes on: [2×3]
- Modes in rectangular wave guide
 - Antenna and its types
